

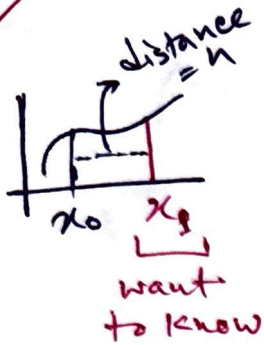
L-7

Chapter - 3.1

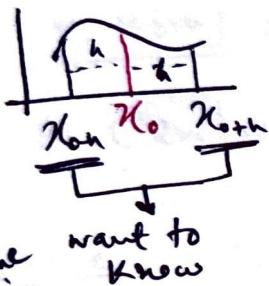
Differentiation

Note!

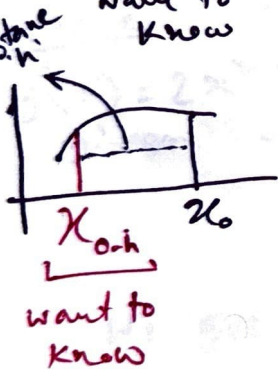
$x_0 \Rightarrow$ is known/given position
 x_{0+h} or, $x_{0-h} \Rightarrow$ is desired/need to calculate position.



$$\Rightarrow \text{Forward differentiation} = \underbrace{\frac{f(x_0+h) - f(x_0)}{h}}_{\text{Approx. calculated value}} + \underbrace{\left| \frac{f''(\xi)(-h)}{2} \right|}_{\text{truncation error}}$$



$$\Rightarrow \text{Central Differentiation} = \underbrace{\frac{f(x_0+h) - f(x_0-h)}{2h}}_{\text{Approx. calculated value}} - \underbrace{\left| \frac{f'''(\xi)(h^2)}{3!} \right|}_{\text{truncation Error}}$$



$$\Rightarrow \text{Backward differentiation} = \frac{f(x_0) - f(x_{0-h})}{h}$$

Here, ' h ' is step size. Means, the distance between the given position (x_0) & calculated/want to know position (x_{0+h}).

Keynote: If you reduce h size, you will get more accurate result.

$$h = 0.1 \text{ (good)}$$

$$h = 0.01 \text{ (better)}$$

$$h = 0.0001 \text{ (may give more correct value)}$$

Let's check an Example

$$f(x) = 2x^2 \quad ; \quad h = 0.1; \quad \text{want to know } f'(2) = ??$$

using actual dif.

$$\frac{d}{dx} f(x) = \frac{d}{dx} (2x^2) = 2 \cdot 2x \Rightarrow 2 \cdot (2 \times 2) \Rightarrow \boxed{8}$$

Now using our value calculation formula

$$\boxed{\text{Center dif.}} \Rightarrow f'(x) = \frac{f(x_0+h) - f(x_0-h)}{2h}$$
$$= \frac{2x(2+0.1)^2 - 2x(2-0.1)^2}{2 \times 0.1}$$

Note
 $f(x) = 2x^2$
given

$$\boxed{= 8}$$

We got the same ans! cause short function.

Now, Let's change function, $f(x) = 2x^2 - e^x$
 $h = 0.1$

calculate, $f'(2) = ?$

using actual dif. $\frac{d}{dx} (2x^2 - e^x) \Rightarrow f'(x) = 4x - e^x$

$$\boxed{\therefore f'(2) = 0.6109}$$

using our center dif.

$$f'(x) = \frac{f(x_0+h) - f(x_0-h)}{2h} = \frac{2\{(2+0.1)^2 - e^2\} - 2\{(2-0.1)^2 - e^2\}}{2 \times 0.1}$$

$$\boxed{\therefore f'(2) = 0.5986}$$

Our calculated value is close enough. NOT exactly same.
The difference of value is Truncation Error.

Instead of, $h = 0.1$; If you use $\boxed{h = 0.01}$, check what value you get. Better / Bad ??

Ex. 2:

x	$f(x)$
4	16
4.1	18
4.2	20
4.3	21
4.4	22

$$4.1 - 4 = 0.1$$

$$0.1$$

$$0.1$$

here,
step size
 $h = 0.1$

calculate, $f'(4.2)$

I'm showing using all methods.

Method-1: Using Forward dif

$$f'(4.2) = \frac{f(4.2 + 0.1) - f(4.2)}{2 \times 0.1}$$

$$= \boxed{10}$$

Method-2: Using Backward dif

$$f'(4.2) = \frac{f(4.2) - f(4.2 - 0.1)}{0.1}$$

$$= \boxed{20}$$

Method-3: Using Central dif

$$f'(4.2) = \frac{f(4.2 + 0.1) - f(4.2 - 0.1)}{2 \times 0.1}$$

$$= \frac{21 - 18}{2 \times 0.1}$$

$$= \boxed{15}$$

Let's calculate a truncation error

AKA upper bound error.

$$n = 2 \quad f(x) = \cos(x)$$

$$h = 0.1 \quad x \in [1, 1.5]$$

using Forward Dif.'s Error

$$\left| \frac{f''(\xi) (-h)}{2} \right|$$
$$= |- \cos(1)| \times \left| \frac{-0.1}{2} \right|$$

$$= 0.54 \times 0.05$$

$$\boxed{= 0.027}$$

Here

$$f(x) = \cos(x)$$
$$f'(x) = -\sin(x)$$
$$f''(x) = -\cos(x)$$

Now $x \in [1, 1.5]$

gives $\leftarrow \cos(1) = 0.54$
max value